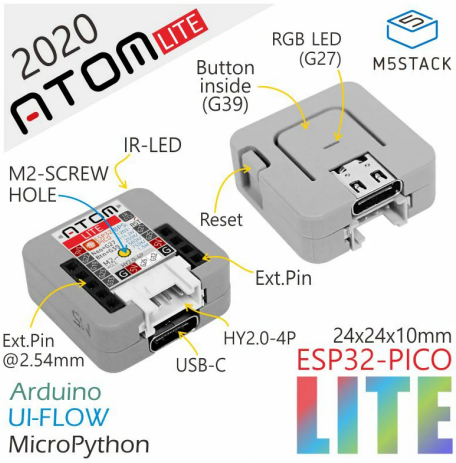
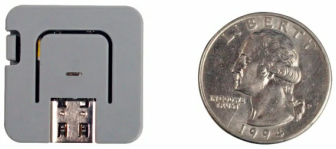
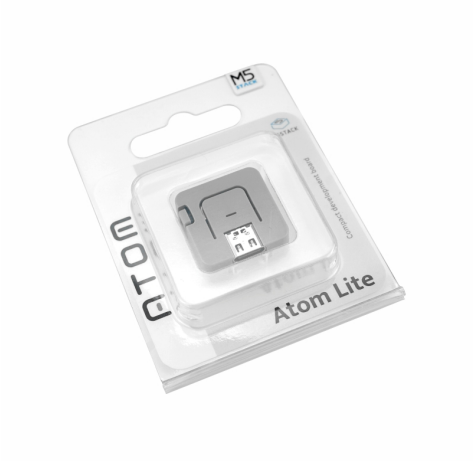
Atom-Lite

SKU:C008









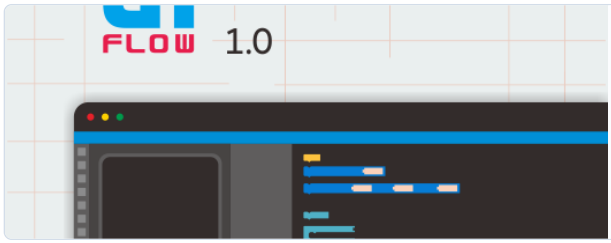




Description

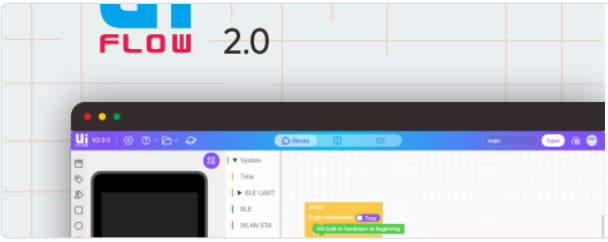
Atom-Lite is a very compact development board in the M5Stack development kit series, with a size of **only 24.0 x 24.0 mm**, providing more GPIO for user customization, making it very suitable for embedded smart hardware development. The main controller uses the **ESP32-PICO-D4** solution, integrates a **Wi-Fi** module, has a built-in 3D antenna, and features **4 MB** of SPI flash, providing Infra-Red, RGB Led, button, and GROVE/HY2.0 interfaces. The onboard USB Type-C interface allows for quick program upload and download, and there is an M2 screw hole on the back for fixing.

Tutorial



UiFlow1

This tutorial will show you how to control the Atom-Lite device using the UiFlow1 graphical programming platform.



UiFlow2

This tutorial will show you how to control the Atom-Lite device using the UiFlow2 graphical programming platform.

Features

- Based on ESP32 development
- Compact body
- Built-in infrared transmission function
- Programmable button
- RGB LED indicator
- Expandable pins and interfaces
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Atom-Lite

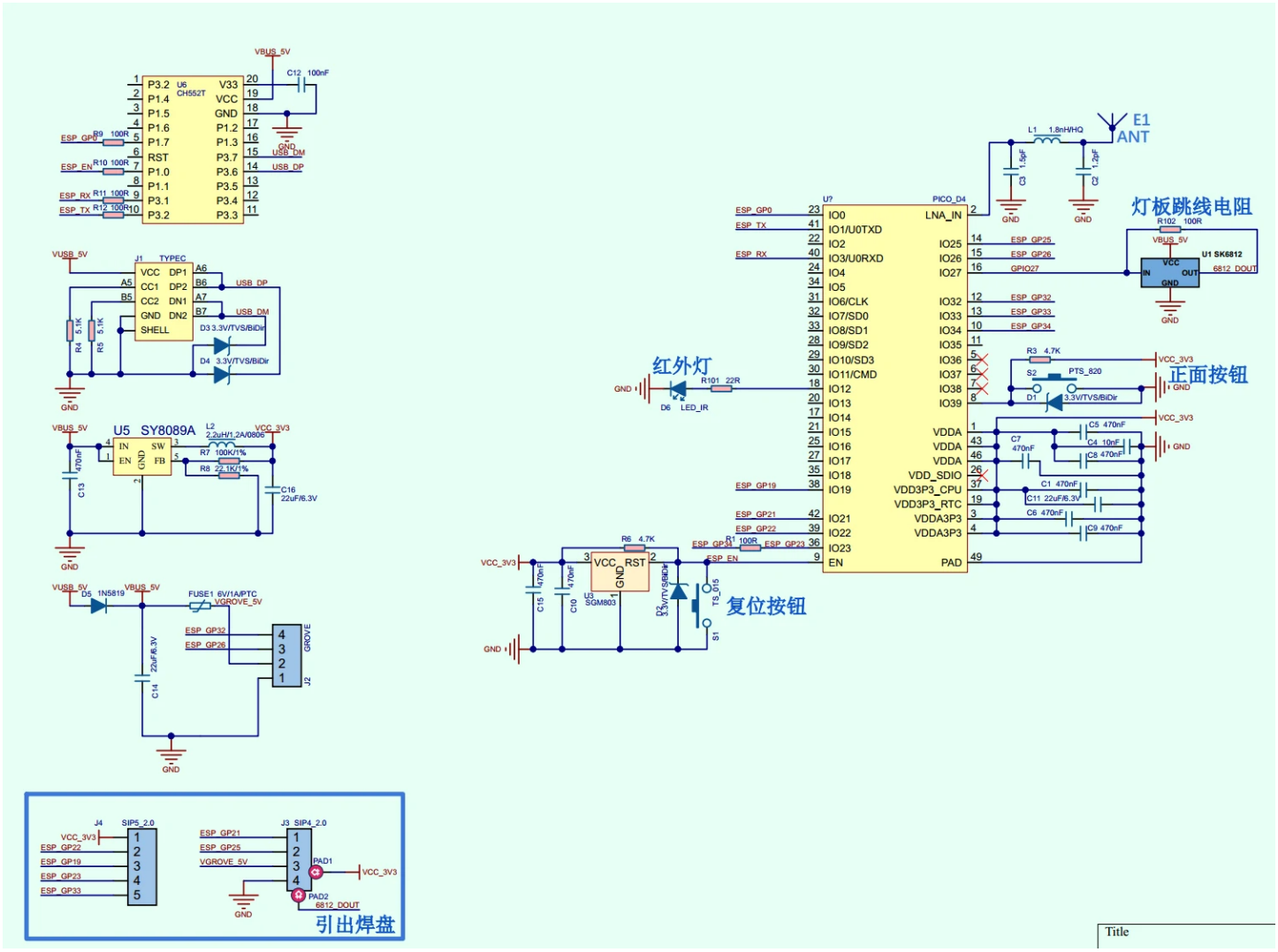
Applications

- IoT nodes
- Microcontrollers
- Wearable devices

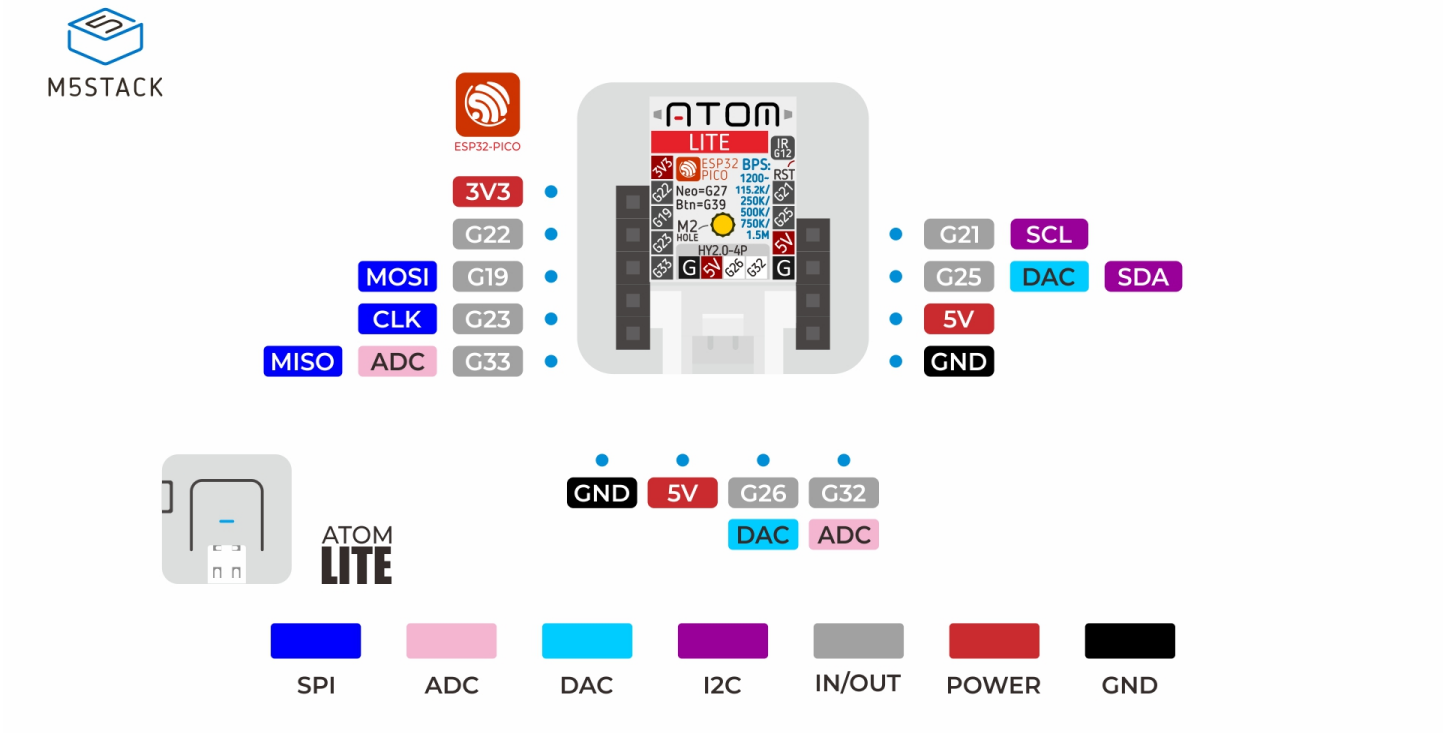
Specifications

Specification	Parameter
SoC	ESP32-PICO-D4 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency up to 240MHz
DMIPS	600
SRAM	520KB
Flash	4MB
Wi-Fi	2.4 GHz Wi-Fi
Input Voltage	5V @ 500mA
Host Interface	USB Type-C x 1, GROVE(I2C+I/O+UART) x 1
PIN Interface	G19, G21, G22, G23, G25, G33
RGB LED	SK6812 3535 x 1
IR	IR transmitter
Button	Customizable button x 1
Antenna	2.4G 3D antenna
Operating Temp	0 ~ 40°C
Case Material	Plastic (PC) + ABS
Product Size	24.0 x 24.0 x 9.5mm
Product Weight	5.5g
Package Size	81.0 x 65.0 x 13.0mm
Gross Weight	10.9g

Schematics


灯板跳线电阻
红外灯
正面按钮
复位按钮
引出焊盘
Title

PinMap


M5STACK
ESP32-PICO
3V3
G22
MOSI
G19
CLK
G23
MISO
ADC
G33
G21
SCL
G25
DAC
SDA
5V
GND
ATOM LITE
GND
5V
G26
G32
DAC
ADC
SPI
ADC
DAC
I2C
IN/OUT
POWER
GND

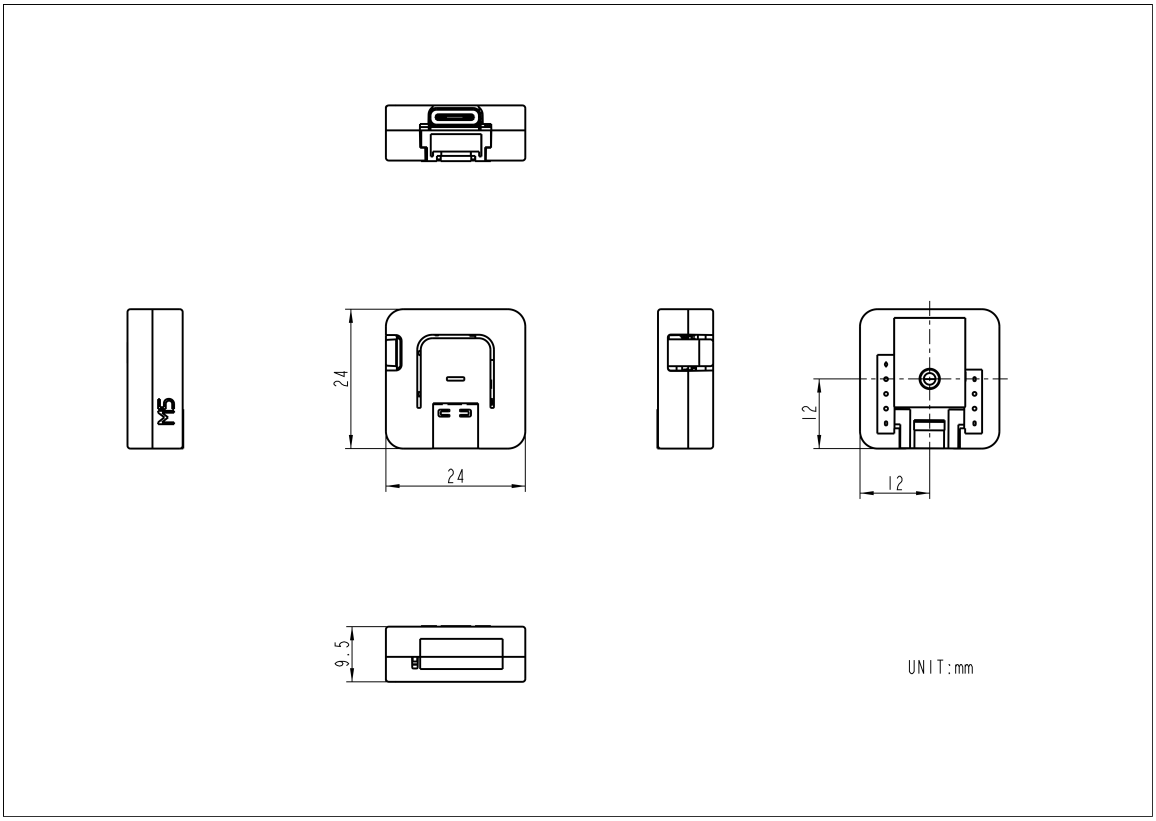
RGB & Button & IR & I2C

ESP32-PICO-D4	G27	G39	G12	G21	G25
RGB	RGB				
Button		Button			
IR			IR		
I2C				SCL	SDA

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G26	G32

Model Size



Structure

- ATOM Lite Structure Files

Datasheets

- ESP32-PICO

Softwares

Arduino

- Atom Arduino Library

UiFlow1

- Atom-Lite UiFlow1 Quick Start
- Atom-Lite Test Example

UiFlow2

- Atom-Lite UiFlow2 Quick Start
- Atom-Lite Test Example

Home Assistant

- Home Assistant Integration

PlatformIO

```
[env:m5stack-atom]
platform = espressif32@6.7.0
board = m5stack-atom
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

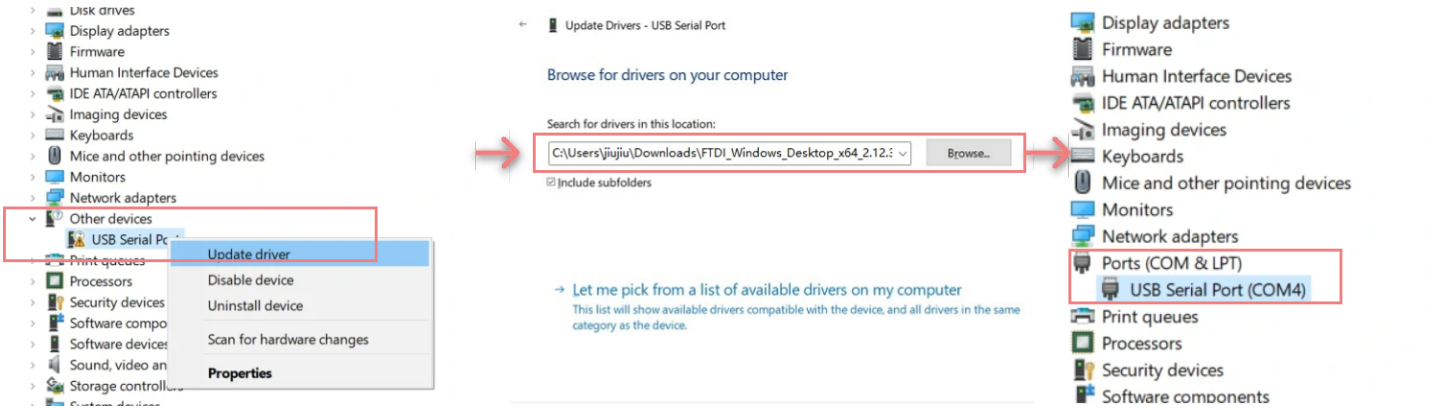
USB Driver

Atom-Lite may not work without a driver on some systems. You can resolve this issue by manually installing the FTDI VCP driver.

1. Select the driver file that matches your operating system from the list below, download it, and extract it. You can also visit the FTDI Chip official website for additional FTDI driver versions.

Driver Name	Download Link
Windows_Desktop_x64	Download
Windows_Desktop_x86_32	Download
Windows_Desktop_ARM	Download
MacOS_Intel	Download
MacOS_ARM64	Download

2. Connect the device to your PC, open Device Manager, and install the FTDI driver (the screenshot below uses Windows 10 as an example). (Note: On some systems, you may need to install the driver twice for it to take effect. The unrecognized device is usually named `M5Stack` or `USB Serial`.)



Easyloader

Easyloader	Download Link	Note
Atom-Lite Factory Test Easyloader	download	/

Video

Product Comparison

To compare information on the Atom series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.